ASSIGNMENT - II

II. Change of Scale Property, First Shift Theorem, Inverse Laplace Transforms of Standard of Logarithmic Functions and Inverse Functions:

1. If $L^{-1}\left[\frac{e^{-1/s}}{\sqrt{s}}\right] = \frac{\cos 2\sqrt{t}}{\sqrt{\pi t}}$ show that $L^{-1}\left[\frac{e^{-a/s}}{\sqrt{s}}\right] = \frac{\cos 2\sqrt{at}}{\sqrt{\pi t}}$, where $a > 0$

2. Find $L^{-1}\left[\frac{s^2}{(s-2)^3}\right]$

   Ans : $e^{2t} + 4te^{2t} + 2e^{2t}t^2$

3. Find $L^{-1}\left[\frac{14s+10}{49s^2+28s+13}\right]$

   Ans : $\frac{2}{7}e^{-(2/7)t}\left\{\cos\frac{3}{7}t + \sin\frac{3}{7}t\right\}$

4. Find $L^{-1}\left[\log\left(\frac{s^2-1}{s^2}\right)\right]$

   Ans : $\frac{2}{t}[1 - \cosh t]$

5. Find $L^{-1}\left[\tan^{-1}\left(\frac{2}{s}\right)\right]$

   Ans : $\frac{\sin 2t}{t}$